Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 · STANDARD 6.DS.1

Understand Statistical Questions

Lesson 2-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

FRONT OFFICE MISSION

Scout the Hardwood

You are the new data analyst for the Riverside Hoops youth league. The head coach drops a stack of questions on your desk — some will yield rich data full of variety, others just one fixed fact. Master statistical questions and you decide what the front office actually investigates.

My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can tell the difference between a statistical question and a non-statistical question.

Language Objective: I can explain my reasoning using the words statistical question, data, variability, and survey.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Statistical Question | Non-statistical Question | Data | Variability | Survey | Data distribution | Dot plot

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A question that expects a variety of answers and is answered by collecting data is a _____.

2. A question that does not anticipate variability in the data is a _____.

3. Facts or numbers collected to answer a question are _____.

4. How much the values in a data set differ from one another is the _____.

5 A way to collect data by asking many people the same question is a _____.

6 The way data values are spread out across a range is the _____.

7 A number line that shows one dot for each data value is a _____.

Write About the Math

Which question is statistical? Why would the editor's question give the newspaper more interesting data?
¿Qué pregunta es estadística? ¿Por qué la pregunta de la editora le daría al periódico datos más interesantes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Statistical Question**
  Pregunta estadística
- ☐ **Non-statistical Question**
  Pregunta no estadística
- ☐ **Data**
  Datos
- ☐ **Variability**
  Variabilidad
- ☐ **Survey**
  Encuesta

Model: *A statistical question expects answers that vary across many cases.* — Students identify "How tall is each player?" as statistical because heights vary, while "How tall is our point guard?" has one fixed answer.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The reporter's question is ____ because ____. The editor's question is ____ because ____. The editor's question is better for a story because ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.
Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Statistical Question
2. **Notes 2:** Non-statistical Question
3. **Notes 3:** Data
4. **Notes 4:** Variability
5. **Notes 5:** Survey
6. **Notes 6:** Data distribution
7. **Notes 7:** Dot plot
8. **Watch for this mistake:** *A common mistake in Statistical Questions and Data is judging a question by its topic or its numbers instead of its variability. Students see a question about ONE player, ONE game, or ONE fixed fact — like "How many points does the team need to win a game?" — and call it statistical just because it has a number or is about sports. Or they see a question about a single player's whole season, like "How many touchdowns did the quarterback throw this season?", and call it NOT statistical because only one player is named, forgetting that a season covers many games with a different result each time. Both errors skip the real test: will the answers actually differ from case to case (game to game, player to player)? If yes, it is a statistical question; if there is one fixed answer, it is not.*
9. **Try It 1:** C. How many rebounds does each player grab per game? — *"How many rebounds does each player grab per game?" anticipates variability — different players grab different numbers of rebounds, so you must collect data. The other three each have one fixed answer.*
10. **Try It 2:** B. Because the warm-up times will vary from one athlete to another — *A statistical question anticipates variability. Different athletes warm up for different amounts of time, so the data you collect will vary.*